

Khent Benedict L. Balulot

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Professional Summary

Aspiring Network and Cloud Engineer with strong foundations in computer engineering, networking, and embedded systems. Experienced in configuring enterprise-grade switches, cloud infrastructure on AWS, and developing IoT-based automation projects. Skilled in Python, Linux, virtualization, and system administration. Recognized as a DOST-SEI Scholar (Top 6.56% Exam score nationally) and former CHED CMSP Scholar, with proven hands-on experience in network troubleshooting, embedded electronics, and cloud engineering.

Beyond technical expertise, competed internationally as a HipHop dancer across Asia and the USA — demonstrating discipline, creativity, and the ability to balance academics, professional work, and personal passions while consistently delivering quality results.

Technical Skills

Programming & Development:

- Languages & Frameworks:** C/C++, Python, JavaScript (Vanilla JS, Three.js), HTML5, CSS3, Flask (Web Framework), Flask-SocketIO
- Tools & Libraries:** Pandas, NumPy, Socket.IO, Web Speech API, Three.js (3D Visualization), TailwindCSS
- Version Control:** Git/GitHub
- Database Systems:** MySQL, JSON (localStorage), Google Sheets API

Web Development & Real-Time Systems:

- Frameworks:** Flask Web Framework (Backend APIs, Template Rendering), Flask-SocketIO (WebSocket Integration)
- Protocols & Technologies:** WebSockets (Socket.IO), REST APIs, Event-Driven Architecture
- Real-Time Systems:** Real-time Multi-Device Synchronization, Bidirectional Communication, State Broadcasting, Editor/Viewer Role Systems, Live 3D Visualization Sync

Networking & Infrastructure:

- Core Protocols:** TCP/IP, DNS, DHCP, Routing Protocols, BGP, QoS
- Switching & Routing:** VLAN Configuration (Segmentation, Security), Port Mirroring (Deep Packet Inspection), Enterprise Switch Configuration (Avaya, Extreme Networks)
- Monitoring & Security:** SNMPv3, IPFIX (Traffic Analysis), Syslog, Network Security Hardening (Telnet/HTTP Disable, SSH Enable), Firewall Rules
- Application Layer:** VoIP (Voice over IP), VPN Overlay Networks

Linux & Cloud:

- Platforms:** Linux CLI (Bash), Apache Web Server, AWS (EC2, EBS, S3, VPC), GCP (Basics), MySQL, Roundcube Email Server, Kali Linux
- Practices:** Bash scripting for system automation and deployment, AWS EC2 instance management (t3.micro, t3.medium) with EBS snapshots and S3 backup solutions, MySQL database administration, backup, and migration, CTF challenges on [Kali Linux](#) (penetration testing tools, enumeration, vulnerability scanning)

Embedded Systems & Electronics:

- Microcontrollers & Platforms:** ESP32, ESP32-C3, Arduino, LattePanda SBC
- Tools & Libraries:** Pandas, NumPy, Socket.IO, Web Speech API, Three.js (3D Visualization), TailwindCSS, Git/GitHub
- Database Systems:** MySQL, JSON (localStorage), Google Sheets API

Systems & Administration:

- Virtualization & Orchestration:** Proxmox VE (nested virtualization, container management), Docker (containerization)
- Networking & Firewall:** MikroTik RouterOS (WAN/LAN segmentation, firewall rules, DHCP), Access Control Lists (ACLs), Firewall Rule Configuration, VPN Overlay Networks
- Directory & Identity Services:** Windows Active Directory, Tailscale (secure remote access)

Next Goal (Data Science & AI):

- Core Skills:** Machine Learning, Deep Learning, Generative AI, LLMs (GPT, Gemini, LLaMA, Falcon,DeepSeek), Natural Language Processing (NLP), Time Series Analysis, Model Deployment, Prompt Engineering.
- Frameworks & Tools:** LangChain, RAG (Retrieval-Augmented Generation), LlamaIndex, LangGraph, LangGraph, PhiData, LangServer, AutoGen, LangSmith, AutoML.
- Applications:** AI-Driven Process Automation, Predictive Modeling, Statistical Analysis, Big Data Technologies, Data Visualization.

Management & Operations:

- Agile Methodologies, Microservices Architecture, Business Analysis, Product Management, Team Leadership, Business Intelligence.

Communication & Collaboration:

- Strong in technical writing, effective communication, stakeholder engagement, and team collaboration, fostering a productive and inclusive environment.

Languages:

- Fluent in English, Tagalog and Hiligaynon.

Experience

Network Engineering Intern | Ubiquity | 2026

- Configured Avaya and Extreme Networks enterprise switches using ASCII scripts and Putty.
- Provisioned VLANs, IP addressing, and gateway assignments across 50+ ports.
- Implemented 7 VLANs (2, 3, 9, 42, 56, 252, 253) for traffic segmentation and security.
- Configured SNMPv3, IPFIX, and syslog for monitoring and traffic analysis.
- Set up port mirroring for deep packet inspection with Gigamon and Intouch recorder.
- Hardened switch security by disabling Telnet/HTTP and enabling SSH.

- Collaborated with senior engineers to validate production-ready configurations.

Cloud Engineering Intern | Ubiquity | 2026

- Observed and assisted in AWS server management and deployment.
- Gained hands-on familiarity with EC2 instances (t3.micro, t3.medium), snapshots, and scaling.
- Optimized EBS storage and configured S3 for backup solutions.
- Assisted in VPC setup for private/public networking.
- Practiced MySQL commands for database management.
- Supported migration of Roundcube email databases from production to staging for optimization.

IT Technician & Network Troubleshooter | Family-Owned Computer Shop | Ongoing

- Serviced 50+ devices with hardware repair and OS troubleshooting.
- Configured local networks, routers, DNS/DHCP for small businesses.
- Provided customer support and diagnostics.

Student Assistant | Culture and Arts Office, CHMSU | 2024–2026

- Supported administrative and event operations while maintaining full-time academic standing.

Thesis Commission Contractor | Multiple Universities | 2023–Present

- Designed and delivered embedded systems projects (PCB layout, firmware, sensors).
- Completed 6+ successful commissions for BS Engineering theses.

Projects

KYUN AI - Web-Enabled Voice-Activated 6-Axis Multimodal Robotic Arm (2026)

Developed a full-stack web-enabled 6-axis robotic arm with Flask-SocketIO WebSocket server enabling real-time multi-device synchronization and Editor/Viewer role-based control. Integrated Three.js 3D visualization with click-and-drag manipulation, PID motor control with configurable Kp/Ki/Kd parameters, and Web Speech API for voice commands with dual Professional/Sassy personality modes. Implemented Serial communication protocol (0xFF header + 6 angle bytes) for Arduino servo control, LED On/Off/Blink, and absolute stepper positioning (0-360°), alongside Teach Pendant recording, 12-slot memory system with smooth execution transitions, collision zone limits, and real-time analytics dashboard with cross-client sync.

Safety House IoT System (2025)

Architected an ESP32-based home safety system with Telegram Bot API and SIM800 GSM module for redundant SMS alerts achieving 98% notification efficiency. Designed a multi-sensor network monitoring gas leaks, temperature anomalies, and motion detection with a low-latency real-time web dashboard for continuous monitoring. Implemented non-blocking sensor polling with priority-based interrupt handling ensuring critical alerts bypass queue processing for instant response.

ESP32 Dual-Node Priority Alert & GPS Tracker (2025)

Built a two-node IoT system using ESP32 master and ESP32-C3 slave communicating via ESP-NOW protocol with non-blocking priority-based loop architecture where interrupts preempt sensor polling for instant alert delivery. Integrated ultrasonic distance sensor (100 cm threshold), water-level ADC calibration, vibration motor, and buzzer with dynamic web dashboards on both nodes and GPS tracking forwarding coordinates to Google Sheets at configurable intervals. Optimized resource usage for concurrent sensing, web UI serving, and priority event handling on constrained MCUs.

Automated Farm Irrigation System (2019)

Developed a self-sustaining automated irrigation system as a self-taught high school student using multiple soil moisture sensors and environmental data without AI assistance. Implemented control logic with relay-actuated water pumps optimizing resource efficiency through hardcoded decision-making based on real-time soil conditions.

Data Center in a Box (2024)

Deployed a Proxmox VE nested virtualization lab with MikroTik WAN/LAN segmentation and custom firewall rules integrating Alpine Linux, Windows XP (DHCP-assigned IP), MikroTik CHR router, and Pi-hole LXC container with blocklist configuration. Configured Tailscale secure remote access and custom IP/VLAN routing architecture for isolated network segmentation, demonstrating comprehensive virtualization, routing, and DNS filtering expertise.

Microcontroller-Based Projects (2024-2026) - Automated Rice Fodder Machine, Automated Sulfur Detector with SMS Alerts, Solar Paddle Wheel Water Quality Monitor

Designed complete system architecture, PCB layouts (KiCad/EasyEDA), and firmware for three production-ready embedded systems deployed in real-world conditions. Calibrated sensors and optimized resource usage for reliable field performance with SMS alert integration and solar-powered monitoring.

Education

Bachelor of Science in Computer Engineering (2022–Present)

Carlos Hilado Memorial State University (CHMSU)

- DOST-SEI Scholar (JLSS) – Top 6.56% nationally
 - CHED CMSP Scholar
- 3rd Place – ICPEP C++ Programming Competition (NIR)

CERTIFICATIONS

- Google IT Support Professional Certificate
 - Google IT Automation with Python Professional Certificate
 - CCNA Certification Study (Ongoing)
- Google AI Essentials (Planned)
 - Google Advanced Data Analytics (Planned)